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A Physics-Informed Neural-Network Model for Rechargeable Thick-Electrode Lithium-Metal Battery Cells that is Accurate at High C-Rate

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***Abstract*—We construct a physics-informed machine learning (PIML) model for rechargeable thick-electrode lithium-metal battery (LMB) cells by combining an enhanced single-particle model (SPMe) with a feedforward neural network (FNN). The hybrid model incorporates an additional pole necessary for modeling variation in electrochemical variables across the thickness of the electrode and guarantees bounded-input bounded-output stability by design. We train the model against the predictions of a Newman model simulated with the PyBaMM package. The trained PIML model predicts the voltage of a cell with a $200\,\mu\mathrm{m}$ electrode at 1.8C with 93% less error than the vanilla SPMe while remaining tractable for embedded applications. We provide Python code to simulate the Newman model, simulate the SPMe, and train the PIML model.**

***Index Terms*—Lithium-metal batteries, physics-informed machine learning, feedforward neural network**

I. Introduction

Rechargeable lithium-metal batteries (LMB) hold promise as successors to lithium-ion batteries (LIB) in weight-critical applications such as class 4–8 electric trucks and buses where energy density as high as $500\,\mathrm{Wh\,kg^{-1}}$ (compared to a maximum of about $265\,\mathrm{Wh\,kg^{-1}}$ in LIB) could enable substantial weight reduction [1]. In the past, LMB technology has remained unexploited due to poor control of lithium (Li) deposition at the Li-metal electrode. The result is formation of a dense layer of inactive (or "dead") Li on the surface of the Li-metal electrode, causing rapid power fade and short cycle life. As advances in material science and pack design overcome these issues [2], [3], [4], [5], [6], [7], it remains to develop tractable models and effective control strategies to make LMB viable for commercial application.

When computational resources are limited, lithium battery cells have been modeled with single-particle models (SPMs) [8], [9], [10], [11]. SPMs are simple enough to embed in applications such as electric vehicles, but lose accuracy at high C-rate due to the assumption that electrochemical variables remain uniform across the thickness of a cell's porous electrodes. This problem is amplified in cells with thick electrodes, which are desirable in weight-critical applications due to their higher energy density [12], [13]. To enable controls that maximize the lifespan and performance of lithium battery cells (e.g., fast-charge controllers), we need models that produce accurate predictions of internal variables (e.g., overpotential) across the thickness of the electrodes while remaining tractable for embedded applications [11], [14], [15], [16], [8].

Recently, researchers have explored combining lithium-battery models with data-driven elements to obtain hybrid models that are simultaneously realizable in embedded applications and accurate at high C-rate [17], [18], [19], [20], [15], [21]. In particular, physics-informed machine learning models (PIMLs)—in which the state of a physical model such as an SPM is used as the input to a data-driven model such as a neural network—enjoy the guarantee of stability, modest training data requirements, and robustness under operating conditions not expressed in the training data [17], [18].

In the same spirit, this article introduces a PIML model for thick-electrode LMB cells combining an enhanced SPM (SPMe) with a feedforward neural network (FNN). We train the hybrid model using the "ground truth" predictions of a Newman model implemented in the Python Battery Mathematical Modeling (PyBaMM) package [22].[1]

Our novel contribution is to introduce an additional pole to the SPMe to improve prediction accuracy at high C-rates. This pole accounts for the variation in lithium concentration along the thickness of the porous electrode that traditional SPMs neglect. Without the pole, we find that the PIML model lacks robustness to operating conditions unseen in the training data.

II. Enhanced Single-Particle Model

To begin, we develop an enhanced SPM (SPMe) for LMB cells from the lumped-parameter partial-differential-equation Newman model given in our previous work [23]. The SPMe models electrolyte diffusion and reaction kinetics in addition to solid diffusion within a representative single particle. We employ the finite-volume method to discretize the diffusion equations in space, which has the advantage of conserving Li mass. As ref. [9] gives detailed derivations of the SPMe equations and a comparison between SPMe's for LMB and LIB cells, only a brief development follows.

[1]We provide Python code for simulating our models at the following URL: https://github.com/whileman133/LMB_PIML_ACC2026

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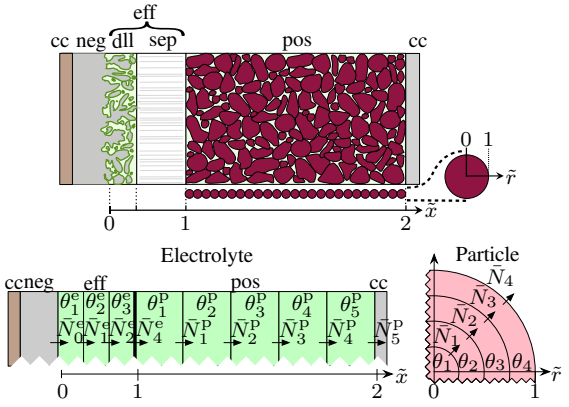


Figure 1: LMB cell sandwich (upper) and spatial discretization (lower).

A. LMB Cell and Coordinate System

Fig. 1 illustrates the regions of an LMB cell: dead-lithium layer (dll), separator (sep), and porous positive electrode (pos) with respective thicknesses L^{d}, L^{s}, and L^{p}. Dimensionless position along the sandwich is $\tilde{x}$ and dimensionless radial position within spherical particles is $\tilde{r}$. We make use of superscripts to denote the locale of the parameters and variables. The "n", "d", "s" and "p" superscripts denote the surface of the Li-metal electrode (neg), dll, sep, and pos, respectively. An "r" superscript is a placeholder for any or all relevant cell regions. The "s" and "e" subscripts denote quantities belonging to the solid and electrolyte phases, respectively. The cell-level voltage is v_{cell} and the applied current is i_{app} ($i_{\mathrm{app}} > 0$ for discharge). As discussed in ref. [23], we assume Li plating/stripping is well-controlled such that dll and sep may be combined into a single "effective" (eff) layer without altering the cell-level voltage response significantly.

B. Spherical Diffusion in the Solid

We model the agglomeration of particles in pos by a single representative spherical particle. In the particle, Fickian diffusion drives mass transport according to:

$$\frac{\partial \theta_{\mathrm{s}}}{\partial t} = -\frac{1}{4\pi\tilde{r}^2}\frac{\partial \bar{N}_{\mathrm{s}}}{\partial \tilde{r}}, \qquad \bar{N}_{\mathrm{s}} \triangleq -\bar{D}_{\mathrm{s}}(\theta_{\mathrm{s}})4\pi\tilde{r}^2\frac{\partial \theta_{\mathrm{s}}}{\partial \tilde{r}},$$

where $\theta_{\mathrm{s}}^{\mathrm{p}} \triangleq c_{\mathrm{s}}^{\mathrm{p}}/c_{\mathrm{s,max}}^{\mathrm{p}}$ is the Li stoichiometry and $\bar{N}_{\mathrm{s}}$ is the stoichiometric Li flux in the solid. Baker and Verbrugge [24] specify the solid diffusion coefficient as:

$$\bar{D}_{\mathrm{s}}^{\mathrm{p}}(\theta_{\mathrm{s}}) = -\bar{D}_{\mathrm{s,ref}}^{\mathrm{p}}\frac{F}{RT}\theta_{\mathrm{s}}\left(1-\theta_{\mathrm{s}}\right)\frac{\partial U_{\mathrm{ocp}}(\theta_{\mathrm{s}})}{\partial \theta_{\mathrm{s}}}. \qquad (1)$$

The boundary conditions are:

$$\bar{N}_{\mathrm{s}}\big|_{\tilde{r}=0} = 0 \qquad \bar{N}_{\mathrm{s}}\big|_{\tilde{r}=1} = 4\pi\frac{|\theta_{100}-\theta_0|}{10,800Q}i_{\mathrm{f}},$$

where i_{f} is the Faradaic Li current directed out of the surface of the particle and Q is the cell's total capacity in Ah. Since we consider a single particle, $i_{\mathrm{f}} = -i_{\mathrm{app}}$.

We discretize these diffusion equations over space by dividing the spherical particle into n concentric spherical shells, as shown in Fig. 1. It can be shown that the time derivative of the Li stoichiometry in each shell is given by:

$$\begin{aligned}
\frac{\partial \theta_1}{\partial t} &= 3\bar{D}_1 n^2\left(\theta_2-\theta_1\right),\\
\frac{\partial \theta_i}{\partial t} &= \frac{3n^2}{i^3-(i-1)^3}\Big[\bar{D}_i i^2\theta_{i+1} - \left(\bar{D}_i i^2 + \bar{D}_{i-1}\left(i-1\right)^2\right)\theta_i\\
&\quad + \bar{D}_{i-1}\left(i-1\right)^2\theta_{i-1}\Big] \quad \forall\, 2 \le i \le n-1,\\
\frac{\partial \theta_n}{\partial t} &= -3\bar{D}_{n-1}n^2\frac{\left(n-1\right)^2\left(\theta_n-\theta_{n-1}\right)}{n^3-\left(n-1\right)^3}\\
&\quad - \frac{n^3}{n^3-\left(n-1\right)^3}\frac{|\theta_{100}-\theta_0|}{3600Q}i_{\mathrm{f}},
\end{aligned}$$

where $\bar{D}_i \triangleq \bar{D}_{\mathrm{s}}(\theta_{\mathrm{s}}(\tilde{r}_i))$ and the shell radii are $\tilde{r}_i = in^{-1}$. In state-space (SS) form, we have:

$$\frac{\mathrm{d}\boldsymbol{x}_{\mathrm{s}}}{\mathrm{d}t} = A_{\mathrm{s}}(t)\boldsymbol{x}_{\mathrm{s}}(t) + B_{\mathrm{s}}i_{\mathrm{app}}(t),$$

where $\boldsymbol{x}_{\mathrm{s}}$ is the column vector of shell stoichiometries, A_{s} is a time-variant tridiagonal matrix, and the B_{s} matrix is:

$$B_{\mathrm{s}} = \left[\mathbf{0}_{1\times(n-1)} \quad \frac{n^3}{n^3-(n-1)^3}\frac{|\theta_{100}-\theta_0|}{3600Q}\right]^T.$$

We employ the backward Euler method to discretize the SS equation in time, giving the discrete-time SS matrices:

$$A_{\mathrm{s,d}}[k] = \left(I_n - t_{\mathrm{s}}A_{\mathrm{s}}(kt_{\mathrm{s}})\right)^{-1}, \quad B_{\mathrm{s,d}}[k] = t_{\mathrm{s}}A_{\mathrm{s,d}}[k]B_{\mathrm{s}},$$

where t_{s} is the sampling interval in seconds.

To simplify the SPMe for embedded applications, we replace the stoichiometry-dependent diffusion coefficient of Eq. (1) by its log-average value over the stoichiometric operating window of the electrode. This makes the discrete-time SS matrices $A_{\mathrm{s,d}}$ and $B_{\mathrm{s,d}}$ time invariant.

C. Linear Diffusion in the Electrolyte

Conservation of anions in the electrolyte requires:

$$\bar{\tau}_{\mathrm{W}}^{\mathrm{r}}\bar{\kappa}^{\mathrm{r}}\bar{\psi}T\frac{\partial \theta_{\mathrm{e}}^{\mathrm{r}}}{\partial t} = -\frac{\partial\left(FN_{\mathrm{e}}^{\mathrm{r}}\right)}{\partial \tilde{x}}, \quad FN_{\mathrm{e}}^{\mathrm{r}} = -\bar{\kappa}^{\mathrm{r}}\bar{\psi}T\frac{\partial \theta_{\mathrm{e}}^{\mathrm{r}}}{\partial \tilde{x}} - i_{\mathrm{e}}^{\mathrm{r}},$$

where $\bar{\kappa}^{\mathrm{r}}$ is the region conductance , $\bar{\tau}_{\mathrm{W}}^{\mathrm{r}}$ is the diffusion time-constant, T is the absolute temperature, $\theta_{\mathrm{e}} \triangleq c_{\mathrm{e}}/c_{\mathrm{e},0}$ is the fractional salt composition, and $N_{\mathrm{e}}^{\mathrm{r}}$ is the anion flux. The boundary conditions are: $N_{\mathrm{e}}^{\mathrm{r}}|_{\tilde{x}=0,3} = 0$, $\theta_{\mathrm{e}}^{\mathrm{r}}|_{\mathrm{left}} = \theta_{\mathrm{e}}^{\mathrm{r}}|_{\mathrm{right}}$, and $N_{\mathrm{e}}^{\mathrm{r}}|_{\mathrm{left}} = N_{\mathrm{e}}^{\mathrm{r}}|_{\mathrm{right}}$.

We discretize these equations over space by dividing eff into n^{e} and pos into n^{p} rectangular prisms as shown in Fig. 1. At the boundary between eff and pos, we define boundary conductance and layer width as follows:

$$\begin{aligned}
\frac{1}{\bar{\kappa}^{\mathrm{ep}}} &\triangleq \frac{\Delta\tilde{x}^{\mathrm{e}}}{\Delta\tilde{x}^{\mathrm{e}}+\Delta\tilde{x}^{\mathrm{p}}}\frac{1}{\bar{\kappa}^{\mathrm{e}}} + \frac{\Delta\tilde{x}^{\mathrm{p}}}{\Delta\tilde{x}^{\mathrm{e}}+\Delta\tilde{x}^{\mathrm{p}}}\frac{1}{\bar{\kappa}^{\mathrm{p}}},\\
\Delta\tilde{x}^{\mathrm{ep}} &\triangleq \frac{\Delta\tilde{x}^{\mathrm{e}}+\Delta\tilde{x}^{\mathrm{p}}}{2},
\end{aligned}$$

where $\Delta\tilde{x}^{\mathrm{r}} = \left(n^{\mathrm{r}}\right)^{-1}$. It can be shown that the time derivative of the salt composition in each prism is given by:

$$\frac{\partial \theta_1^{\mathrm{e}}}{\partial t} = \frac{1}{\bar{\tau}^{\mathrm{e}}}\frac{\theta_2^{\mathrm{e}}-\theta_1^{\mathrm{e}}}{\left(\Delta\tilde{x}^{\mathrm{e}}\right)^2} + \frac{i_{\mathrm{app}}}{\bar{\tau}^{\mathrm{e}}\bar{\kappa}^{\mathrm{e}}\bar{\psi}T\Delta\tilde{x}^{\mathrm{e}}},$$

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$$\frac{\partial \theta_i^{\mathrm{e}}}{\partial t} = \frac{1}{\bar{\tau}^{\mathrm{e}}} \frac{\theta_{i+1}^{\mathrm{e}} - 2\theta_i^{\mathrm{e}} + \theta_{i-1}^{\mathrm{e}}}{\left(\Delta \tilde{x}^{\mathrm{e}}\right)^2} \quad \forall\, 2 \le i \le n^{\mathrm{e}} - 1,$$

$$\frac{\partial \theta_n^{\mathrm{e}}}{\partial t} = \frac{1}{\bar{\tau}^{\mathrm{e}}} \left[\frac{\theta_1^{\mathrm{p}}}{\Delta \xi^{\mathrm{e}} \Delta \tilde{x}^{\mathrm{e}}} - \left\{ \frac{1}{\Delta \xi^{\mathrm{e}} \Delta \tilde{x}^{\mathrm{e}}} + \frac{1}{\left(\Delta \tilde{x}^{\mathrm{e}}\right)^2} \right\} \theta_n^{\mathrm{e}} + \frac{\theta_{n-1}^{\mathrm{e}}}{\left(\Delta \tilde{x}^{\mathrm{e}}\right)^2} \right],$$

in eff and:

$$\frac{\partial \theta_1^{\mathrm{p}}}{\partial t} = \frac{1}{\bar{\tau}^{\mathrm{p}}} \left[\frac{\theta_2^{\mathrm{p}}}{\left(\Delta \tilde{x}^{\mathrm{p}}\right)^2} - \left\{ \frac{1}{\left(\Delta \tilde{x}^{\mathrm{p}}\right)^2} + \frac{1}{\Delta \xi^{\mathrm{p}} \Delta \tilde{x}^{\mathrm{p}}} \right\} \theta_1^{\mathrm{p}} + \frac{\theta_n^{\mathrm{e}}}{\Delta \xi^{\mathrm{p}} \Delta \tilde{x}^{\mathrm{p}}} \right]$$

$$\frac{\partial \theta_i^{\mathrm{p}}}{\partial t} = \frac{1}{\bar{\tau}^{\mathrm{p}}} \frac{\theta_{i+1}^{\mathrm{p}} - 2\theta_i^{\mathrm{p}} + \theta_{i-1}^{\mathrm{p}}}{\left(\Delta \tilde{x}^{\mathrm{p}}\right)^2} \quad \forall\, 2 \le i \le n^{\mathrm{p}} - 1,$$

$$\frac{\partial \theta_n^{\mathrm{p}}}{\partial t} = -\frac{1}{\bar{\tau}^{\mathrm{p}}} \frac{\theta_n^{\mathrm{p}} - \theta_{n-1}^{\mathrm{p}}}{\left(\Delta \tilde{x}^{\mathrm{p}}\right)^2},$$

in pos, where:

$$\Delta \xi^{\mathrm{e}} \triangleq \frac{\bar{\kappa}^{\mathrm{p}} \Delta \tilde{x}^{\mathrm{e}} + \bar{\kappa}^{\mathrm{e}} \Delta \tilde{x}^{\mathrm{p}}}{2 \bar{\kappa}^{\mathrm{p}}}, \quad \Delta \xi^{\mathrm{p}} \triangleq \frac{\bar{\kappa}^{\mathrm{p}} \Delta \tilde{x}^{\mathrm{e}} + \bar{\kappa}^{\mathrm{e}} \Delta \tilde{x}^{\mathrm{p}}}{2 \bar{\kappa}^{\mathrm{e}}}.$$

We write these equations in state-space form as:

$$\frac{\mathrm{d}\boldsymbol{x}_{\mathrm{e}}}{\mathrm{d}t} = A_{\mathrm{e}} \boldsymbol{x}_{\mathrm{e}}(t) + B_{\mathrm{e}} i_{\mathrm{app}}(t),$$

where $\boldsymbol{x}_{\mathrm{e}}$ is the column vector of the salt composition in each prism, A_{e} is a tridiagonal matrix, and the B_{e} matrix is:

$$B_{\mathrm{e}} = \begin{bmatrix} \frac{1}{\bar{\tau}^{\mathrm{e}} \bar{\kappa}^{\mathrm{e}} \bar{\psi} T \Delta \tilde{x}^{\mathrm{e}}} & \mathbf{0}_{1 \times (n^{\mathrm{e}} + n^{\mathrm{p}} - 1)} \end{bmatrix}^T.$$

We discretize this state-space equation in time using the Backward Euler method.

D. Electrolyte Potential

By charge-neutrality assumption of concentrated-solution theory, the point form of Faraday's law gives:

$$\frac{\partial i_{\mathrm{e}}^{\mathrm{r}}}{\partial \tilde{x}} = i_{\mathrm{f+dl}}^{\mathrm{r}}, \qquad i_{\mathrm{e}}^{\mathrm{r}} = -\bar{\kappa}^{\mathrm{r}} \frac{\partial \phi_{\mathrm{e}}^{\mathrm{r}}}{\partial \tilde{x}} + \bar{W} \bar{\kappa}^{\mathrm{r}} \bar{\psi} T \frac{\partial \theta_{\mathrm{e}}^{\mathrm{r}}}{\partial \tilde{x}},$$

where $i_{\mathrm{e}}^{\mathrm{r}}$ is the electrolyte current. In eff, $i_{\mathrm{e}}^{\mathrm{e}} = i_{\mathrm{app}}$. In pos, we assume $i_{\mathrm{f+dl}}^{\mathrm{p}} = -i_{\mathrm{app}}$ such that $i_{\mathrm{e}}^{\mathrm{p}}(\tilde{x}) = (2 - \tilde{x})\, i_{\mathrm{app}}$. By substituting these expressions into the equation for $i_{\mathrm{e}}^{\mathrm{r}}$ above and integrating over $\tilde{x}$, we obtain:

$$\phi_{\mathrm{e}}[k] = -\left(\frac{1}{\bar{\kappa}^{\mathrm{eff}}} + \frac{1}{2\bar{\kappa}^{\mathrm{p}}} \right) i_{\mathrm{app}}[k] + \bar{W} \bar{\psi} T \left(\theta_{\mathrm{e}}[k, 2] - \theta_{\mathrm{e}}[k, 0] \right),$$

where we take the potential reference point as $\phi_{\mathrm{e}}|_{\tilde{x}=0} \triangleq 0$.

E. Current–Overpotential Equations

We assume Butler–Volmer kinetics at the surface of the Li-metal electrode as given by ref. [25]:

$$i_{\mathrm{f}}^{\mathrm{n}} = i_0^{\mathrm{n}} \left\{ \exp\left(\frac{\left(1 - \beta^n\right) F}{RT} \eta_{\mathrm{s}}^{\mathrm{n}} \right) - \exp\left(-\frac{\beta^{\mathrm{n}} F}{RT} \eta_{\mathrm{s}}^{\mathrm{n}} \right) \right\}.$$

We assume Multiple-Site Multiple-Reaction (MSMR) intercalation kinetics at the surface of the porous electrode as given by ref. [26]:

$$i_{\mathrm{f}}^{\mathrm{p}} = \sum_j i_{0,j}^{\mathrm{p}} \left\{ \exp\left(\frac{\left(1 - \beta_j^{\mathrm{p}}\right) F}{RT} \eta_{\mathrm{s}}^{\mathrm{p}} \right) - \exp\left(-\frac{\beta_j^{\mathrm{p}} F}{RT} \eta_{\mathrm{s}}^{\mathrm{p}} \right) \right\},$$

where β^{r} are the symmetry factors of the reduction reactions. The surface overpotential is:

$$\eta_{\mathrm{s}}^{\mathrm{r}} = \phi_{\mathrm{s}}^{\mathrm{r}} - \phi_{\mathrm{e}}^{\mathrm{r}} - U_{\mathrm{ocp,b}}^{\mathrm{r}} - \bar{R}_{\mathrm{f}}^{\mathrm{r}} i_{\mathrm{f+dl}}^{\mathrm{p}},$$

where $U_{\mathrm{ocp,b}}^{\mathrm{r}}$ is the equilibrium potential with respect to the standard Li electrode at the bulk concentrations. For neg, $U_{\mathrm{ocp,b}}^{\mathrm{n}} = 0$, and for pos, $U_{\mathrm{ocp,b}}^{\mathrm{p}} = U_{\mathrm{ocp,b}}^{\mathrm{p}}(\theta_{\mathrm{ss}})$, where θ_{ss} is the Li stoichiometry at the surface of the particle. We employ the MSMR open-circuit-potential (OCP) model given by ref. [26] that consists of $3J$ thermodynamic parameters $\{U_j^0, X_j, \omega_j\}$, where J is the number of intercalate phases. The exchange currents are:

$$i_0^{\mathrm{n}} = \bar{k}_0^{\mathrm{n}} \left(\theta_{\mathrm{s}}^{\mathrm{n}}\right)^{\beta^{\mathrm{n}}} \left(\theta_{\mathrm{e}}^{\mathrm{n}}\right)^{1 - \beta^{\mathrm{n}}},$$

$$i_{0,j}^{\mathrm{p}} = \bar{k}_{0,j} \left(0.5 X_j\right)^{-\omega_j} \left(x_j\right)^{\omega_j \beta_j^{\mathrm{p}}} \left(X_j - x_j\right)^{\omega_j \left(1 - \beta_j^{\mathrm{p}}\right)} \theta_{\mathrm{e}}^{\left(1 - \beta_j^{\mathrm{p}}\right)}.$$

The boundary condition at neg is: $i_{\mathrm{f+dl}}^{\mathrm{n}}\big|_{\tilde{x}=0} = i_{\mathrm{app}}$.

F. Cell Voltage

At time-step k, we compute the cell voltage as the sum:

$$v_{\mathrm{cell}}[k] = \phi_{\mathrm{se}}^{\mathrm{p}}[k] + \phi_{\mathrm{e}}[k] - \phi_{\mathrm{s}}^{\mathrm{n}}[k],$$

where $\phi_{\mathrm{se}}^{\mathrm{p}}$ is the potential difference from the surface of the particle to the electrolyte and $\phi_{\mathrm{s}}^{\mathrm{n}}$ is the potential difference from neg to the electrolyte at $\tilde{x} = 0$. We compute these quantities by inverting the current–overpotential equations assuming $\beta = 0.5$ for all reactions:

$$\phi_{\mathrm{se}}^{\mathrm{p}}[k] = U_{\mathrm{ocp}}(\theta_{\mathrm{ss}}[k]) + \frac{2RT}{F} \sinh^{-1}\left(\frac{i_{\mathrm{f}}^{\mathrm{p}}[k]}{2 i_0^{\mathrm{p}}[k]} \right) + \bar{R}_{\mathrm{f}}^{\mathrm{p}} i_{\mathrm{f}}^{\mathrm{p}}[k],$$

$$\phi_{\mathrm{s}}^{\mathrm{n}}[k] = \frac{2RT}{F} \sinh^{-1}\left(\frac{i_{\mathrm{f}}^{\mathrm{n}}[k]}{2 i_0^{\mathrm{n}}[k]} \right), \; i_0^{\mathrm{p}}[k] \triangleq \sum_j i_{0,j}^{\mathrm{p}}[k].$$

G. Evaluating the Performance of the Vanilla SPMe

Before we hybridize the SPMe with data-driven elements, we compare its predictions to that of the PDE Newman model implemented in PyBaMM. For this purpose, we employ a synthetic LMB cell with a $200\,\mu\mathrm{m}$ LMO positive electrode and known parameter values.[2] We consider the predictions of Newman model as the "ground truth" in evaluating the performance of the SPMe. We implement finite-volume simulations with $n_{\mathrm{s}} = 10$, $n_{\mathrm{e}}^{\mathrm{eff}} = 5$, $n_{\mathrm{e}}^{\mathrm{pos}} = 10$, and $t_{\mathrm{s}} = 1\,\mathrm{s}$.

Fig. 2 plots the surface stoichiometry predicted by the two models for a 1 C constant-current discharge that is interrupted at about $t = 380\,\mathrm{s}$. The upper plot shows the solid-surface stoichiometry at the current-collector edge of the porous electrode ($\tilde{x} = 2$) versus time. The lower plot shows the surface stoichiometry at $t = 380\,\mathrm{s}$ versus position within pos. The predictions of the SPMe differ significantly from the Newman model. The reason for this is that the Newman model accounts for variation of the surface stoichiometry across the thickness of the porous electrode. In the Newman model, after current is interrupted at $t = 380\,\mathrm{s}$, diffusion drives equalization of Li both along the radial dimension within particles ($\tilde{r}$) and between particles in the thickness

[2]The **XLSX_CELLDEFS/cellLMO-P2DM.xlsx** spreadsheet in the Github repository lists the parameter values of the synthetic cell.

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dimension ($\tilde{x}$). These processes manifest as two competing relaxation features in the Newman time response shown in Fig. 2. The time response of the SPMe, however, exhibits only one relaxation feature—that associated with diffusion along the radius of the representative particle—because the SPMe lacks information about the variation along the thickness of the porous electrode. Similar error is present in the electrolyte potential (ϕ_{e}) and faradaic current (i_{f}) predictions produced by the SPMe (*cf.* Fig. 3). Since the cell voltage is a function of θ_{ss}, ϕ_{e}, and i_{f}, these errors propagate to v_{cell}.

Based on the discussion above, it is unreasonable to expect a data-driven model informed solely on the state of an SPM to produce accurate predictions of internal variables and cell voltage; additional dynamic elements are required to model variations in θ_{ss}, ϕ_{e}, and i_{f} across the electrode's thickness. We observe that variation in θ_{ss} across the thickness of the electrode is related to the applied current (i_{app}) by a "leaky integrator" type response. Therefore, as shown in Fig. 4, we apply a single-time-constant filter to $i_{\mathrm{app}}[k]$ to produce an additional state variable $\delta[k]$ that quantifies the variation. We estimate the time constant of the filter from Fig. 2 as $\tau \approx 410\,\mathrm{s}$. As an alternative, we employ orthogonal projection [27] to estimate the mean of the three largest time constants in the $10\,\%$ and $90\,\%$ SOC range ($\tau_1 = 32.95\,\mathrm{s}$, $\tau_2 = 164.4\,\mathrm{s}$, and $\tau_3 = 386.3\,\mathrm{s}$) and construct parallel single-pole filters that produce additional state variables $\delta_1[k]$, $\delta_2[k]$, and $\delta_3[k]$. We refer to these two approaches as the $n_\delta = 1$ and $n_\delta = 3$ formulations, respectively.

III. Physics-Informed Machine Learning Model

Herein, we combine our SPMe with several feedforward neural networks (FNNs) to improve its predictive accuracy. We employ one FNN per variable of interest (θ_{ss}, ϕ_{e}, and i_{f} at $\tilde{x} = 2$). As shown in Fig. 4, the input to each FNN is the state of the SPMe at time step k and the output is an improved estimate of the corresponding variable at time step k ($\hat{\theta}_{\mathrm{ss}}$, $\hat{\phi}_{\mathrm{e}}$, and $\hat{i}_{\mathrm{f}}$). This PIML model structure has a number of advantages, including:

- Stability guarantee: Since feedback is not employed within or around the FNNs, the PIML model is stable provided the SPMe is stable.
- Robustness under novel operating conditions: Unlike a pure data-driven model, the PIML incorporates a physical model more likely to produce accurate predictions when conditions outside of training data are encountered.
- Data efficiency: The FNNs learn only the relationships the physical model fails to describe, reducing training data requirements and the chance of overfitting.

Specifications for each FNN appear in Table I. Input variables common to all FNNs include θ_{ss}, θ_{avg} (average particle stoichiometry), and δ (inhomogeneity); FNN-$\hat{\phi}_{\mathrm{e}}$ and FNN-$\hat{i}_{\mathrm{f}}$ take i_{app} and θ_{e} at $\tilde{x} = 0$ and $\tilde{x} = 2$ as additional inputs. Hidden layers use ReLU activation and output layers use linear activation. As discussed further in Section IV, we find that FNN-$\hat{\phi}_{\mathrm{e}}$ and FNN-$\hat{i}_{\mathrm{f}}$ require additional hidden layers and nodes to achieve the same level of accuracy as FNN-$\hat{\theta}_{\mathrm{ss}}$.

Table I
FNN Specifications

FNN	Inputs from SPMe	n_{h}	w_{h}	N_{p}
FNN-$\hat{\theta}_{\mathrm{ss}}$	θ_{ss}, θ_{avg}, δ	3	32	4,385
FNN-$\hat{\phi}_{\mathrm{e}}$	θ_{ss}, θ_{avg}, δ, $\theta_{\mathrm{e}}(0)$, $\theta_{\mathrm{e}}(2)$, i_{app}	5	128	67,073
FNN-$\hat{i}_{\mathrm{f}}$	θ_{ss}, θ_{avg}, δ, $\theta_{\mathrm{e}}(0)$, $\theta_{\mathrm{e}}(2)$, i_{app}	5	128	67,073

n_{h}: number of hidden layers; w_{h}: number of nodes in hidden layers (layer "width"); N_{p}: number of trainable parameters

We train the weights and biases of each FNN by minimizing the mean-square-error between the output of the FNN and the Newman model at every time-step over several current profiles. We use the Adam optimizer, a batch size of 1024, and 1000 epochs. The following current profiles are used for training:

- Constant current: ±0.2, ±0.4, ±0.6, ±0.8, ±1, ±1.5, ±2 C.
- Galvanostatic intermittent tritiation technique (GITT): 10m/20m, 5m/10m, 2m/5m at ±1C ($t_{\mathrm{d}}/t_{\mathrm{r}}$ where t_{d} is the discharge interval and t_{r} is the rest interval).
- Drive profiles: EPA UDDS and LA92 from 90% to 5% SOC (available from ref. [28]).

For validation, we compute the root-mean-square-error (RMSE) between the predictions of the FNNs and the Newman model for the following current profiles:

- Constant current: ±0.5, ±1.8 C.
- GITT: 7m/14m at ±1.5C.
- Drive profile: EPA US06 from 90% to 5% SOC [28].

We employ the same SPMe parameter values as used in Section II-G and a sampling interval of $t_{\mathrm{s}} = 1\,\mathrm{s}$.

Note that this model training procedure may be generalized from the synthetic cell to a real cell by substituting parameter values identified from laboratory experiments. Ref. [29] introduces experiments for this purpose.

IV. Results and Analysis

Table II lists the RMSE observed between the trained PIML and Newman models for θ_{ss}, ϕ_{e}, and i_{f} at $\tilde{x} = 2$ for all discharge-current profiles used in training and validation. In addition, we list the RMSE between the Newman model and cell voltage computed from the outputs of the PIML model. For comparison, we list the same RMSE values for the vanilla SPMe without FNNs. Fig. 3 compares predictions of the vanilla SPMe, PIML models with $n_\delta = 0, 1, 3$, and the Newman model graphically for several discharge current-profiles used in validation.

The predictive accuracy of the PIML model with $\delta[k]$ input is substantially higher than the vanilla SPMe in all training and validation cases. This is true especially at higher C-rates. We note that the accuracy of the PIML model is sensitive to the width of the FNNs. For FNN-$\hat{\theta}_{\mathrm{ss}}$, little improvement in accuracy occurs above 32 hidden nodes; for FNN-$\hat{\phi}_{\mathrm{e}}$ and FNN-$\hat{i}_{\mathrm{f}}$, the same is true above 128 nodes. Note that we purposely exclude i_{app} as an input to FNN-$\hat{\theta}_{\mathrm{ss}}$, as we have prior knowledge that direct feedthrough from i_{app} to θ_{ss} does

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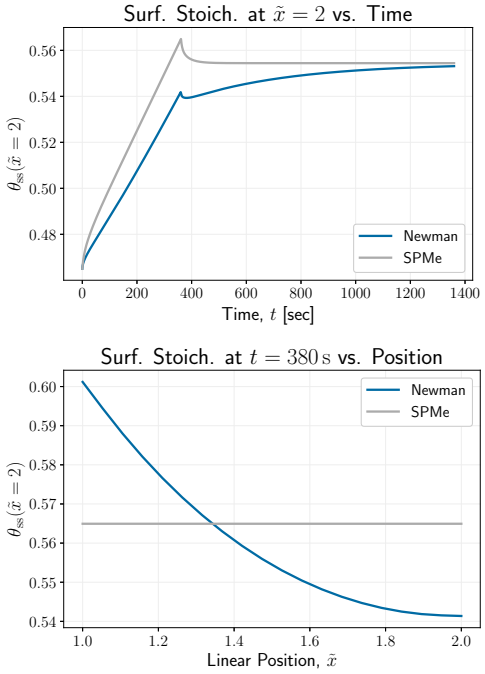


Figure 2: Comparing the solid-surface stoichiometry predicted by the Newman model to that predicted by the SPMe for 1C discharge that is interrupted at about 380s.

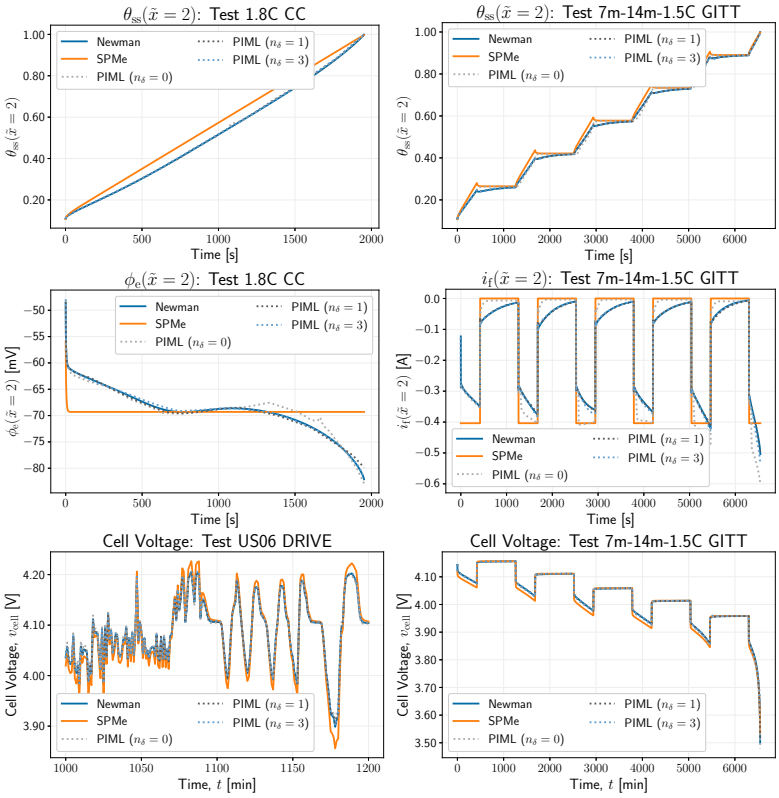


Figure 3: Comparing model predictions of θ_{ss}, ϕ_e, i_f, and v_{cell}.

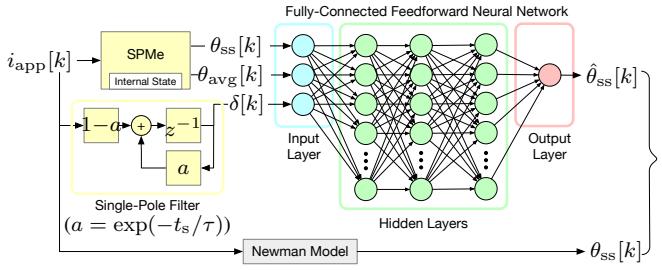


Figure 4: PIML model for predicting $\theta^{p}_{ss}(\tilde{x} = 2)$.

not exist; we observe erroneous "jumps" in θ_{ss} when i_{app} is employed as an additional input to FNN-$\hat{\theta}_{ss}$.

Without the electrode-variation input $\delta[k]$, the predictive accuracy of the PIML model is still higher than the vanilla SPMe. However, closer inspection of Fig. 3 reveals that the model is "confused" by novel operating conditions (i.e., inputs not expressed in the training data), as indicated by substantial prediction error of FNN-$\hat{\phi}_e$ for 1.8C discharge and FNN-$\hat{\theta}_{ss}$/FNN-$\hat{i}_f$ for GITT discharge. This supports our claim that FNNs informed on the state of an SPMe alone cannot make accurate predictions of quantities that vary across the thickness of the porous electrode.

The accuracy of the PIML model with $n_\delta = 3$ is slightly worse than the $n_\delta = 1$ model for the validation profiles, which we attribute to overfitting. A single time-constant appears to be sufficient to model variation along the thickness of the electrode for the studied cell.

Finally, we compare the wall-clock execution time (WCT) of the vanilla SPMe with the hybrid PIML model on a PC with fixed resources. The mean WCT per time-step over all of the training and validation profiles is 38 μs for the vanilla SPMe and 70 μs for the hybrid PIML model. Although some slowdown is apparent, we believe the hybrid model is suitable for embedded applications, particularly when deployed on devices with dedicated neural processing units.

V. Summary and Further Work

In this paper, we developed a PIML model for thick-electrode LMB cells by combining an SPMe with FNNs to correct the predictions of electrochemical variables. The SPMe was augmented with a single-pole filter to model variations along the thickness of porous electrodes that traditional SPMs neglect. We trained the hybrid model against the predictions of a lumped-parameter PDE Newman model implemented in the PyBaMM package. The accuracy of the trained PIML model is comparable to the PDE model—even at high C-rate—but the PIML model has the advantage that it is simple enough to embed in applications such as electric vehicles.

The training method presented here requires a fully-parameterized PDE model generated from laboratory experiments using the methods presented by ref. [29] (for example). As this may be impractical or inconvenient, developing a method to train hybrid PIML models from cell-level measurements is an area for further research.

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Table II
RMS PREDICTION ERROR OF THE SPME AND PIML MODELS

			Vanilla SPMe				PIML Model ($n_\delta = 0$)				PIML Model ($n_\delta = 1$)				PIML Model ($n_\delta = 3$)			
	Profile	Specs.	$\theta_{ss}(2)$ [milli]	$\phi_e(2)$ [mV]	$i_f(2)$ [mA]	v_{cell} [mV]	$\theta_{ss}(2)$ [milli]	$\phi_e(2)$ [mV]	$i_f(2)$ [mA]	v_{cell} [mV]	$\theta_{ss}(2)$ [milli]	$\phi_e(2)$ [mV]	$i_f(2)$ [mA]	v_{cell} [mV]	$\theta_{ss}(2)$ [milli]	$\phi_e(2)$ [mV]	$i_f(2)$ [mA]	v_{cell} [mV]
Training	CC	0.2C	6.08	0.11	1.95	**1.92**	1.07	0.03	2.56	**0.62**	0.21	0.01	0.17	**0.14**	0.11	0.01	0.18	**0.08**
		0.4C	11.99	0.34	6.25	**3.86**	2.16	0.05	2.65	**0.98**	0.25	0.02	0.17	**0.19**	0.13	0.04	0.36	**0.07**
		0.6C	17.68	0.68	12.42	**5.80**	4.22	0.10	2.64	**1.96**	0.31	0.02	0.27	**0.21**	0.14	0.06	0.54	**0.13**
		0.8C	23.07	1.10	20.17	**7.72**	6.58	0.17	3.52	**2.77**	0.32	0.02	0.25	**0.30**	0.13	0.06	0.79	**0.20**
		1C	28.29	1.59	29.31	**9.66**	9.21	1.09	30.11	**0.75**	0.27	0.02	0.31	**0.25**	0.16	0.07	0.90	**0.25**
		1.5C	40.23	3.09	57.41	**14.48**	5.10	0.50	2.89	**3.37**	0.39	0.03	0.73	**0.74**	0.24	0.07	1.59	**0.78**
		2.0C	50.77	4.90	91.32	**19.35**	2.85	0.58	3.55	**2.03**	0.38	0.05	1.15	**0.78**	0.19	0.08	2.60	**1.22**
	GITT	10/20m/1C	13.33	1.89	34.46	**6.53**	6.21	1.14	17.40	**1.23**	0.35	0.04	0.56	**0.21**	0.17	0.05	0.67	**0.20**
		5/10m/1C	11.53	2.03	36.81	**6.51**	6.37	1.15	16.60	**1.36**	0.28	0.03	0.54	**0.28**	0.17	0.05	0.60	**0.13**
		2/5m/1C	9.33	1.98	36.01	**6.06**	5.48	0.93	12.92	**1.34**	0.33	0.04	0.54	**0.25**	0.19	0.05	0.60	**0.15**
	UDDS	90–5%	6.03	1.93	40.23	**5.51**	4.83	0.34	7.90	**1.61**	0.49	0.08	0.74	**0.20**	0.47	0.08	0.93	**0.18**
	LA92	90–5%	9.90	2.86	61.08	**8.38**	7.25	0.45	8.48	**2.28**	0.67	0.12	1.24	**0.29**	0.59	0.12	1.24	**0.26**
Testing	CC	0.5C	14.87	0.50	9.12	**4.84**	3.16	1.28	5.43	**1.28**	0.29	0.08	1.37	**0.33**	0.13	0.08	0.91	**0.14**
		1.8C	46.72	4.14	77.17	**17.40**	3.57	0.89	100.15	**7.38**	2.18	0.34	8.58	**1.20**	2.78	0.21	2.53	**4.13**
	GITT	7/14m/1.5C	18.20	2.99	54.37	**9.79**	11.76	2.74	43.35	**3.43**	0.64	0.25	2.39	**0.38**	1.47	0.23	5.98	**1.11**
	US06	90–5%	23.53	4.77	102.63	**14.95**	14.07	2.09	40.67	**6.50**	1.18	0.71	9.80	**1.53**	2.69	0.85	9.12	**2.26**

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